@qa_insights

Top 15 Jenkins CI/CD

Interview Questions & Answers

Beginner → Intermediate → Advanced (Scenario Based)

What You'll Learn

- ✓ Core Jenkins Concepts
- ✓ Pipelines & Jenkinsfile
- ✓ Plugins & Integrations
- ✓ Real-world Scenarios
- ✓ Best Practices

Perfect For

- ★ Freshers
- ★ QA Testers
- ★ Automation Testers
- ★ DevOps Engineers
- ★ Anyone Preparing for CI/CD Interviews

Real-Time QA_Insights Examples

- ✓ qa_insights Pipelines
- ✓ qa_insights Repositories
- ✓ qa_insights Reports
- ✓ qa_insights Deployments
- ✓ End-to-End CI/CD Flow

Save this post for your next interview!
Swipe Left to explore all 15 questions →

Top 15 Jenkins CI/CD Interview Questions & Answers

1 What is Jenkins?

Answer :

Jenkins is an open-source automation server used to automate building, testing, and deploying applications. It helps implement Continuous Integration (CI) and Continuous Delivery/Deployment (CD), allowing developers and testers to detect issues early and deliver software faster.

Key Points :

- ★ Open Source
- ★ CI/CD Tool
- ★ Automates Build
- ★ Runs Automated Tests
- ★ Supports Plugins
- ★ Cross Platform

QA_Insights Example :

Repository:
qa_insights-ecommerce

Pipeline Flow:

Developer Pushes Code
↓
Jenkins Build Starts
↓
Run Selenium Tests
↓
Generate Allure Report
↓
Deploy to QA Server

2 What is CI/CD?

Answer :

Continuous Integration (CI)
automatically builds and tests code whenever developers commit changes.

Continuous Delivery (CD)
prepares the application for release automatically.

Continuous Deployment (CD)
automatically deploys every successful build into production without manual approval.

Key Points :

- ✓ Frequent Code Merge
- ✓ Automated Testing
- ✓ Faster Feedback
- ✓ Less Manual Work
- ✓ Quick Releases
- ✓ Better Quality

QA_Insights Example :

Developer commits code
↓
qa_insights Git Repository
↓
Jenkins Pipeline
↓
Smoke Test
↓
Deploy to QA

3 Why do we use Jenkins?

Answer :

Jenkins eliminates repetitive manual tasks by automating build, testing, deployment, report generation, notifications, and scheduling.
It improves software quality, reduces human errors, and speeds up release cycles.

Key Points :

- ✓ Automation
- ✓ Scheduling
- ✓ Notifications
- ✓ Reports
- ✓ Integrations
- ✓ Faster Releases

QA_Insights Example :

Instead of manually executing:

Build Project
↓
Run Selenium Suite
↓
Generate Report
↓
Send Email

Jenkins performs everything automatically after every Git commit.

Top 15 Jenkins CI/CD Interview Questions & Answers

4 What is a Jenkins Pipeline?

Answer :

A Jenkins Pipeline is a series of automated stages that define how software moves from source code to deployment. Pipelines are written using Jenkinsfile and support complex automation workflows.

Key Points :

- ★ Stages
- ★ Steps
- ★ Jenkinsfile
- ★ Automation
- ★ Version Controlled
- ★ Reusable
- ★ Scalable

QA_Insights Example :

qa_insights Pipeline Flow

Checkout Code
↓
Build Project
↓
Run Unit Tests
↓
Run Selenium Tests
↓
Publish Allure Report
↓
Deploy to QA Environment

5 What is a Jenkinsfile?

Answer :

A Jenkinsfile is a text file stored inside the project repository that defines the pipeline as code.

It makes CI/CD version-controlled, reusable, and easy to maintain.

Key Points :

- ✓ Pipeline as Code
- ✓ Stored in Git
- ✓ Reusable
- ✓ Version Controlled
- ✓ Easy Maintenance
- ✓ Shareable
- ✓ Supports Best Practices

QA_Insights Example :

Sample Jenkinsfile (qa_insights)

```
pipeline {
   agent any
   stages {
      stage('Build') {
         steps { echo 'Building project' }
      }
      stage('Test') {
         steps { echo 'Running tests' }
      }
      stage('Deploy') {
         steps { echo 'Deploying to QA' }
      }
   }
}
Repository: qa_insights-demo-project
```

6 Difference between Declarative and Scripted Pipeline?

Answer :

Declarative Pipeline uses a structured syntax and is easier for beginners.

Scripted Pipeline uses Groovy programming and provides greater flexibility for complex workflows.

Key Points :

Declarative Pipeline
- ✓ Structured Syntax
- ✓ Easy to Write
- ✓ Readable
- ✓ Recommended for Beginners
- ✓ Less Code

Scripted Pipeline
- ✓ Groovy Based
- ✓ More Flexible
- ✓ Supports Complex Logic
- ✓ Advanced Use Cases
- ✓ More Control

QA_Insights Example :

Declarative Pipeline (qa_insights)

Checkout → Build → Test → Report → Deploy

Scripted Pipeline (qa_insights)

Code → Groovy Logic → Build → Test → Report

Deploy

Top 15 Jenkins CI/CD Interview Questions & Answers

7 What are Jenkins Nodes, Master and Agent?

Answer :

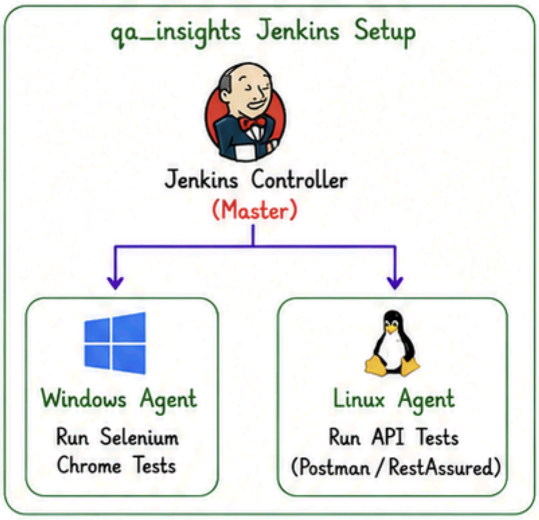
The Jenkins Controller (formerly Master) manages jobs, configurations, and scheduling. Agents (Nodes) are remote machines that execute the actual builds and test cases.

This architecture helps in distributed builds and better scalability.

Key Points :

- ★ Controller (Master)
- ★ Agent (Node)
- ★ Distributed Builds
- ★ Scalability
- ★ Parallel Execution
- ★ Load Distribution

QA_Insights Example :



8 What are Jenkins Plugins?

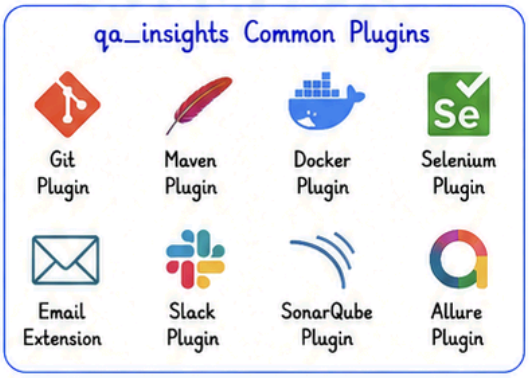
Answer :

Plugins extend Jenkins functionality by integrating third-party tools and technologies. They help in SCM integration, build tools, notifications, testing, code quality analysis, containerization, and more.

Key Points :

- ✓ Extend Functionality
- ✓ Easy Integration
- ✓ Wide Range of Plugins
- ✓ Community Support
- ✓ Regular Updates
- ✓ Improve Productivity

QA_Insights Example :



9 How do you trigger Jenkins Jobs?

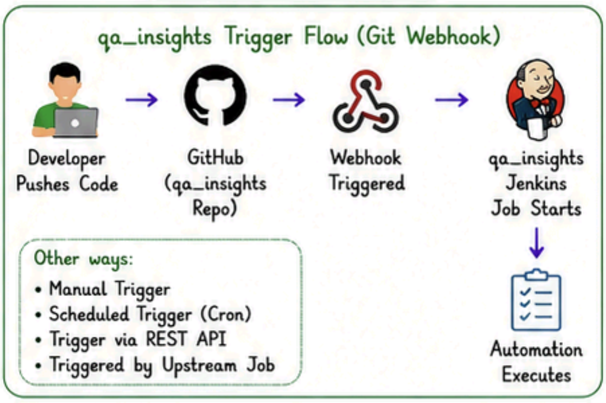
Answer :

Jenkins jobs can be triggered in multiple ways depending on the requirement. Automation is key to CI/CD, so we mostly use automatic triggers like Git webhooks, schedules, APIs, or upstream/downstream jobs.

Key Points :

- ◆ Manual Trigger
- ◆ Git Webhook
- ◆ Scheduled (Cron)
- ◆ API Trigger
- ◆ Upstream Job
- ◆ Event Based Trigger

QA_Insights Example :



Top 15 Jenkins CI/CD Interview Questions & Answers

10 Explain your Jenkins Pipeline in your current project.

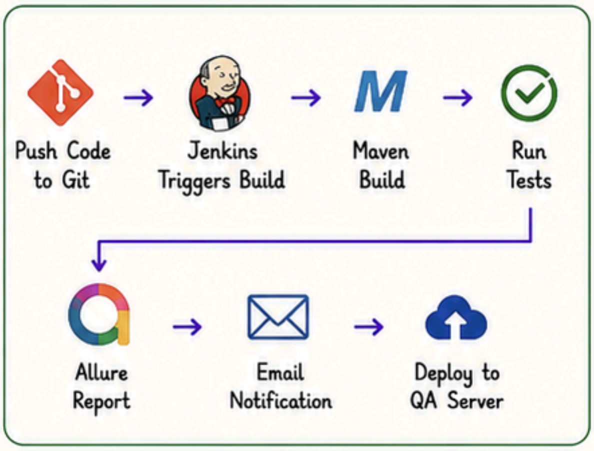
Answer :

In my current project (qa_insights), whenever developers push code to the Git repository, Jenkins automatically checks out the code, builds the project using Maven, runs Selenium regression suites, generates Allure reports, sends email notifications, and deploys to the QA environment if all tests pass.

Key Points :

- Git Integration
- Maven Build
- Selenium Automation
- Allure Reports
- Email Notifications
- Deployment to QA

QA_Insights Pipeline Flow :



11 How do you run Selenium tests in Jenkins?

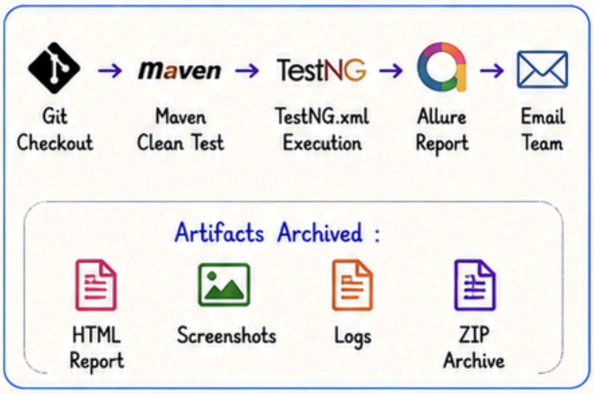
Answer :

I configure the Jenkins job, pull the code from Git, install dependencies using Maven, execute the TestNG suite, generate Allure reports, and archive the artifacts for future reference.

Key Points :

- Job Configuration
- Git Checkout
- Dependency Setup (Maven)
- Execute TestNG Suite
- Generate Allure Report
- Archive Artifacts
- Email Team

QA_Insights Example Flow :



12 How do you handle failed builds?

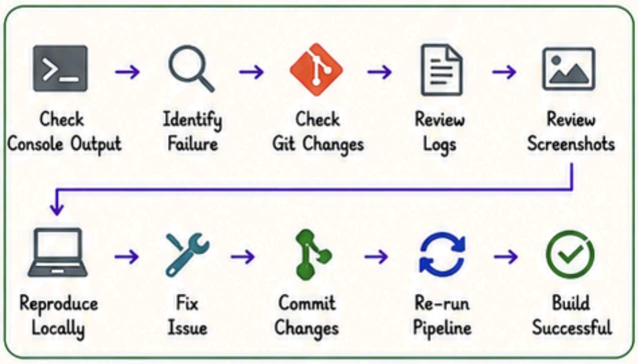
Answer :

I first review the Jenkins Console Output, identify the build or test failures, check Git changes, review logs and screenshots, reproduce the issue locally if needed, fix it, commit the changes, and rerun the pipeline.

Key Points :

- Review Console Logs
- Analyze Failure
- Check Git Changes
- Review Logs/Screenshots
- Reproduce Locally
- Fix and Commit
- Re-run Pipeline
- Verify Success

QA_Insights Failure Handling Flow :



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13 Scenario:

The Jenkins build is successful, but Selenium tests fail. What will you do?

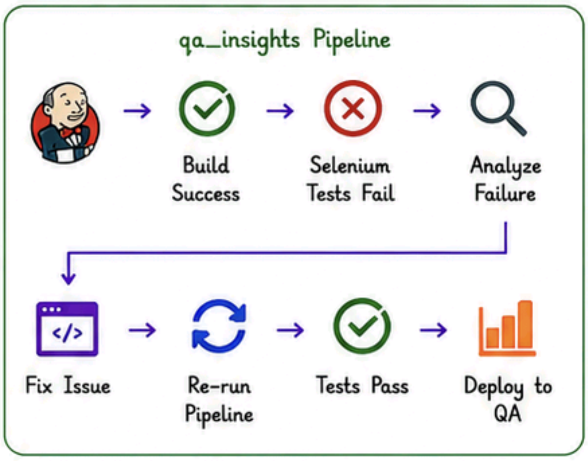
Answer :

I analyze the failed test cases, review logs and screenshots, verify application changes, check environment stability, inspect locators and waits, reproduce the issue locally, and identify whether it's an application defect or automation issue before rerunning.

Key Points :

- ★ Review Logs
- ★ Check Screenshots
- ★ Validate Environment
- ★ Check Locators & Waits
- ★ Reproduce Locally
- ★ Identify Root Cause
- ★ Fix & Re-run

QA_Insights Scenario Flow :



14 Scenario:

Your Jenkins pipeline suddenly stops working after code changes. What will you do?

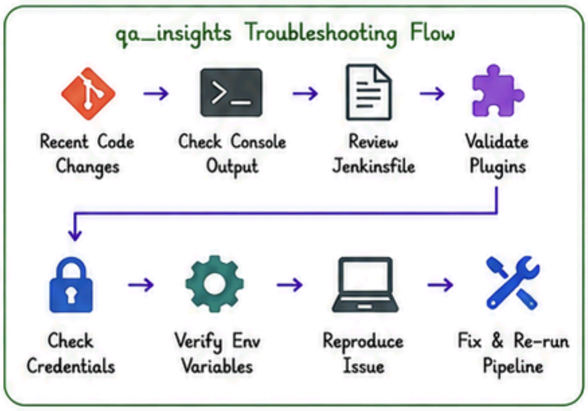
Answer :

I verify the recent Git commits, inspect the console output, check Jenkinsfile changes, validate plugin versions, confirm credentials and environment variables, and reproduce the issue before fixing and rerunning the pipeline.

Key Points :

- ✔ Check Git History
- ✔ Inspect Console Output
- ✔ Review Jenkinsfile
- ✔ Validate Plugins
- ✔ Check Credentials
- ✔ Verify Environment Variables
- ✔ Reproduce & Fix
- ✔ Re-run Pipeline

QA_Insights Scenario Flow :



15 Scenario:

How do you execute automation on multiple browsers simultaneously?

Answer :

I configure Jenkins parallel stages or multiple agents to run tests on Chrome, Firefox, and Edge at the same time. This reduces execution time and ensures cross-browser compatibility.

Key Points :

- ★ Parallel Execution
- ★ Multiple Browsers
- ★ Multiple Agents
- ★ Reduce Execution Time
- ★ Cross-Browser Testing
- ★ Merge Reports
- ★ Better Coverage

QA_Insights Example Flow :